Answer 4a:

$$\forall n \in Z, 3 \mid (3n)$$

$$That\ is, for\ every\ integer\ n, 3n\ is\ divisible\ by\ 3.$$

Answer 4b:

Let n be any arbitrary integer (this is the **generic particular**).

After 3 birthdays, the total amount received is:

$$Total = n + n + n = 3n$$

To prove that this amount is divisible by 3, we use the definition of divisibility:

A number is divisible by 3 if it can be written in the form $3k, where\ k \in Z$.

Now,

$$3n = 3 \times n$$

$Let\ k = n\ where\ k \in Z$

$Thus,$

$$3n = 3k$$

Hence, 3n is divisible by 3.

Conclusion:

For any integer n, the total amount after 3 birthdays is divisible by 3.

Question 5

a.

Step 1: Change variable

Let $j = i - 1 \Rightarrow i = j + 1$

Step 2: Change limits

- When $i = n \Rightarrow j = n - 1$
- When $i = 2n \Rightarrow j = 2n - 1$

Step 3: Substitute

$$n - i + 1 = n - (j + 1) + 1 = n - j$$

$$n + i = n + (j + 1) = n + j + 1$$

Final Answer:

$$\prod_{j=n-1}^{2n-1} \frac{n - j}{n + j + 1}$$

b:

Step 1: Change variable

Let $j = i - 1 \Rightarrow i = j + 1$

Step 2: Change limits

- When $i = 1 \Rightarrow j = 0$
- When $i = n - 1 \Rightarrow j = n - 2$

Step 3: Substitute

$$\frac{i}{(n-1)^2} = \frac{j+1}{(n-1)^2}$$

Final Answer:

$$\sum_{j=0}^{n-2} \frac{j+1}{(n-1)^2}$$

Question 6:

Base Case (n = 2):

LHS:

$$\sum_{i=1}^{1} i(i+1) = 1(2) = 2$$

RHS:

$$\frac{2(1)(3)}{3} = 2$$

LHS=RHS

Inductive Hypothesis:

Assume true for $n = k$:

$$\sum_{i=1}^{k-1} i(i+1) = \frac{k(k-1)(k+1)}{3}$$

Inductive Step (n = k+1):

$$\sum_{i=1}^{k} i(i+1) = \sum_{i=1}^{k-1} i(i+1) + k(k+1)$$

Using hypothesis:

$$= \frac{k(k-1)(k+1)}{3} + k(k+1)$$

Factor:

$$= k(k+1)\left(\frac{k-1}{3} + 1\right)$$

$$= k(k+1)\left(\frac{k+2}{3}\right) = \frac{k(k+1)(k+2)}{3}$$

Which equals RHS for n=k+1

Conclusion: True for all n≥2

Question 7

a

Explicit Formula:

$$g_n = 1 + 2^n$$

b

Suppose that $g_1, g_2, g_3, \ldots$ is a sequence defined as follows:

$$g_1 = 3, g_2 = 5, g_k = 3g_{k-1} - 2g_{k-2} \text{ for every integer } k \geq 3.$$

Prove that $g_n = 2^n + 1$ for every integer $n \geq 1$.

Proof. Let the property $P(n)$ be the equation $g_n = 2^n + 1$.

Show that $P(1)$ and $P(2)$ are true:

We must show that $g_1 = 2^1 + 1$ and $g_2 = 2^2 + 1$. The left-hand side of the first equation is 3 (by definition of $g_1, g_2, g_3, \ldots$), and its right-hand side is $2^1 + 1 = 2 + 1 = 3$ also. The left-hand side of the second equation is 5 (by definition of $g_1, g_2, g_3, \ldots$), and its right-hand side is $2^2 + 1 = 4 + 1 = 5$ also. Thus $P(1)$ and $P(2)$ are true.

Show that for every integer $k \geq 2$, if $P(i)$ is true for each integer i with $1 \leq i \leq k$, then $P(k+1)$ is also true:

Let k be any integer with $k \geq 2$ and suppose that $g_i = 2^i + 1$ for each integer i with $1 \leq i \leq k$. *[This is the inductive hypothesis.]* We must show that $g_{k+1} = 2^{k+1} + 1$. Now

$$\begin{aligned}
g_{k+1} &= 3g_k - 2g_{k-1} && \text{by definition of } g_1, g_2, g_3, \ldots \\
&= 3(2^k + 1) - 2(2^{k-1} + 1) && \text{by inductive hypothesis} \\
&= 3 \cdot 2^k + 3 - 2^k - 2 && \text{by distributing} \\
&= 2 \cdot 2^k + 1 && \text{by combining like terms} \\
&= 2^{k+1} + 1 && \text{by laws of exponents}
\end{aligned}$$

[as was to be shown]. □

Question 8:

[Pre-condition: m is a nonnegative integer, x is a real number, $i = 0$, and $exp = 1$.]

while $(i \neq m)$
  1. $exp := exp \cdot x$
  2. $i := i + 1$

end while

[Post-condition: $exp = x^m$]

loop invariant: $I(n)$ is "$exp = x^n$ and $i = n$."

Proof. [The wording of this proof is almost the same as that of Example 5.5.2.]

I. Basis Property: *[$I(0)$ is true before the first iteration of the loop.]*

$I(0)$ is "$exp = x_0$ and $i = 0$." According to the pre-condition, before the first iteration of the loop $exp = 1$ and $i = 0$. Since $x_0 = 1$, $I(0)$ is evidently true.

II. Inductive Property: *[If $G \wedge I(k)$ is true before a loop iteration (where $k \geq 0$), then $I(k+1)$ is true after the loop iteration.]*

Suppose k is any nonnegative integer such that $G \wedge I(k)$ is true before an iteration of the loop. Then as execution reaches the top of the loop, $i \neq m, exp = x^k$, and $i = k$. Since $i \neq m$, the guard is passed and statement 1 is executed. Now before execution of statement 1, $exp_{\text{old}} = x^k$, so execution of statement 1 has the following effect:

$$exp_{\text{new}} = exp_{\text{old}} \cdot x = x^k \cdot x = x^{k+1}.$$

Similarly, before statement 2 is executed, $i_{\text{old}} = k$, so after execution of statement 2,

$$i_{\text{new}} = i_{\text{old}} + 1 = k + 1.$$

Hence after the loop iteration, the two statements $exp = x^{k+1}$ and $i = k + 1$ are true, and so $I(k+1)$ is true.

III. Eventual Falsity of Guard: *[After a finite number of iterations of the loop, G becomes false.]* The guard G is the condition $i \neq m$, and m is a nonnegative integer. By I and II, it is known that for every integer $n \geq 0$, if the loop is iterated n times, then $exp = x^n$ and $i = n$. So after m iterations of the loop, $i = m$. Thus G becomes false after m iterations of the loop.

IV. Correctness of the Post-Condition: *[If N is the least number of iterations after which G is false and $I(N)$ is true, then the value of the algorithm variables will be as specified in the post-condition of the loop.]* According to the post-condition, the value of exp after execution of the loop should be x^m. But when G is false, $i = m$. And when $I(N)$ is true, $i = N$ and $exp = x^N$. Since both conditions (G false and $I(N)$ true) are satisfied, $m = i = N$ and $exp = x^m$, as required. $\square$